



QUALITY SYSTEM

FORM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 262080

Phase I

Section A – Test Details

Initiator:	Cuong Du (Written by Qiyue Chen)	Test Date:	01-Sep-2026	Date Initiated:	08-Sep-2026	
Date of Incident:	08-Sep-2026	Test Name:	Celsis Sterility	Sample ID:	ETX-260828-0527	
Name of Analyst who Performed the Test:	Andrew Carrillo (AC) (Celsis Sterility Prepper) Gabrielle Surber (GS) (Celsis Sterility Processor) Cuong Du (CCD) (Celsis Sterility Aliquoting & Reading Analyst) America Alanis (ALA) (Sample Coordination & Data Entry)	Sample / Active Name	MOTs-C 10 MG/ML (5 ML) Injection	Lot #:	LG342010349	
		Dosage Form	Injection			
Description of Incident:	On 08 Sep 2026, sample ETX-260828-0527 was found to be positive for growth after Celsis Sterility Testing ☒ Client Care notified of OOS ☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS					
	SOP / Test Method # MICRO-SOP-44	Effective Date: 03-Jan-2025	SOP / Test Method Rev: 01	Limits/Specification: Pass/Fail		
Manager notified	Manager/designee name: Kathan Parikh	Date: 08-Sep-2026	Quality Assurance Department notified:	Yes ☒ No ☐		

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy			Comments	
Analyst interviewed?	☒ Yes ☐ No ☐ N/A		Yes, analysts Andrew Carrillo, Gabrielle Surber, America Alanis & Cuong Du were comprehensively interviewed.	
Correct samples analyzed?	☒ Yes ☐ No ☐ N/A		Yes, Sample ID: ETX-260828-0527	
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A		Yes, as per MICRO-SOP-44	
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A		Yes, as per MICRO-SOP-44	
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A		Yes, Information is available in Eagle	
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A		Yes, all analysts are trained and	
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A		Not Applicable	
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A		Not Applicable	
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A		Yes, the samples were stored	
Material Discrepancy			Lot Number	Expiry Date
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A			
Was the sample transported appropriately?	☒ Yes ☐ No ☒ N/A			
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A			



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Materials and Reagents TSB: 06242026-1 FTM: 06162026-2 PBS (Phosphate Buffered Saline): 35725002	Materials and Reagents TSB: 22 Sep 2026 FTM: 14 Sep 2026 PBS: 31 Dec 2027 Bovine Serum Albumin (BSA): 23 Jan 2028
Standard Used	Celsis ATP Positive Control	022601-1483	31 Jan 2027
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	E001736 (114),	Dec 2026, Jun
System Performance Check (if applicable)?	☒ Yes ☐ No ☐ N/A	E002222	Sep 2026
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☐ N/A	See Phase I	See Phase I
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☐ N/A	See Phase I	See Phase I
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001356 (Monitored by Sensor E001450) Incubator E001357 (Monitored by Sensor E001449)	Jan 2027 Feb 2027 Jan 2027 Feb 2027 Aug 2027 Feb 2027 Aug 2027

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Are chromatogram/spectra/raw data correct?				☐ Yes	☐ No	☒ N/A	Not Applicable	Not Applicable
Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)								
☒ N/A		☐ Re-vialing of original working sample solution						
☐ Re-injection of sample vial		☐ Re-dilution of a working sample solution from the original stock						
☐ Re-injection of original standard vial		☐ Re-dilution of a working standard solution from the original stock standard						
☐ Re-vialing of original working standard solution		☐ Other: N/A QYC 08Sep2026						
Section D – Phase I Summary								
Phase I Summary: All analysts involved in the prepping, processing, aliquoting, and reading of the samples – Andrew Carrillo (AC), Gabrielle Surber (GS), America Alanis (ALA) and Cuong Du (CCD) were interviewed comprehensively. Their answers are recorded throughout this document. Upon arrival, the sample was stored in accordance with the Client's instructions. Both the sample prepping analyst - Andrew Carrillo (AC) - and the processing analyst - Gabrielle Surber (GS) - verified the sample's integrity throughout both the preparation and processing stages. No leaks or turbidity were observed at any point, verifying the integrity of the sample. All reagents and supplies mentioned in the material section above were stored according to the suppliers' recommendations, and their integrity was visually verified before utilization. Moreover, all reagents and supplies had valid expiration dates. The functionality of all equipment was confirmed by reviewing data generated by our comprehensive in-house continuous monitoring system. The sample was processed within the ISO 5 Biological Safety Cabinet (BSC), E001316, situated in the innermost cleanroom (114B), which connects to the intermediate ISO 7 buffer room (114A). This buffer room, in turn, leads to the outermost ISO 8 Anteroom (114). Furthermore, a positive air pressure system is maintained, ensuring airflow from the innermost ISO 7 cleanroom to the intermediate ISO 7 buffer room and finally to the outermost ISO 8 Anteroom. The innermost ISO 7 cleanroom 114B and its ISO 5 BSC E001316 were thoroughly cleaned and prepared before initiating the testing by analyst Gabrielle Surber as per MICRO-SOP-2 (Environmental Monitoring of the Clean Room Facility), and MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). Similarly, for Celsis aliquoting, the ISO 7 cleanroom 114A and its ISO 5 BSC E001798 were thoroughly cleaned and prepared by analyst Cuong Du before initiating the testing as per MICRO-SOP-44 and MICRO-SOP-9. Both the BSCs, E001316 in 114B and E001798 in 114A, were certified and approved by the Engineering and Quality Assurance teams prior to use.								

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Phase I Summary (continued):

On 01 Sep 2026, the sample vials for ETX-260828-0527 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each sample vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. The secondary disinfection happened in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before being transferred to the innermost ISO 7 cleanroom (114B). Inside 114B, the processing analyst, Gabrielle Surber, performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316. Inside the BSC, the vials were placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44 (Celsis Sterility Testing). Direct inoculation was performed by adding 25 mL per media (5 x 300mL FTM jars and 5 x 300mL TSB jars; 1g of Bovine Serum Albumin in 100mL of PBS and 5mL added per suitability method 2601120497). Following testing, the media bottles were subsequently transferred into designated incubators, E001356 and E001357, to initiate incubation.

Upon completion of incubation on 08 Sep 2026, TSB & FTM bottles for ETX-260828-0527 were disinfected and transferred to the intermediate ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In 114A, the media bottles were disinfected one more time before transferring them to the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by analyst Cuong Du (CCD). After aliquoting, Celsis Sterility Reading was performed in accordance with MICRO-SOP-44 on Celsis instrument E002222 by analyst Cuong Du (CCD).

Following the reading, sample ETX-260828-0527 was found to yield positive readings in two of the 300mL FTM media jars (jars #3 and #5). The average Relative Luminescence Units (RLU) from the duplicate reading tubes, originating from the FTM sample jars, yielded 7190 RLU, which exceeded the negative control cutoff (FTM negative control = 985 RLU), confirming a positive result. The TSB media jars tested negative with an RLU of 1500 (TSB cutoff = 2007 RLU, negative control = 669 RLU). All other FTM and TSB sample bottles in the same batch tested negative. The %CV from the duplicate reading tubes was within specification (< 30%). Additionally, all Daily Controls, including the Instrument Blank, Reagent Blank, and ATP Positive Control, were within the defined specifications, each with a %CV below 30%.

Following the OOS result, the positive FTM bottles for ETX-260828-0527 were submitted for Differential Staining and Microbial Identification under ETX-260908-0593. No microbial growth was recovered from the subculture (0 CFU on plate, see Table 1), and differential staining was inconclusive as no cell morphology could be visualized, indicating that the microorganism present in the sample was potentially viable but not culturable.

The culture media utilized were within their expiry period. The negative culture media bottles for the direct inoculation method for the original culture were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative control.

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for Cleanroom 114 was performed on 30 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). During cleaning, it was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom 114.

Tables 2 & 3 (please see attached) present the environmental monitoring results for the duration of testing. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30-35°C and no less than an additional five days at 20-25°C as per MICRO-SOP-2 (Environmental Monitoring of the Cleanroom Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

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Phase I Summary (continued):

Environmental Monitoring from Celsis Sterility Processing:

A review of the Environmental Monitoring (EM) results for the relevant testing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC (E001316) surface sampling plates on the date of testing (01Sep2026), the preceding sampling date (31Aug2026), or the subsequent sampling date (02Sep2026).

Environmental Monitoring from Celsis Sterility Aliquoting:

After reviewing the Environmental Monitoring (EM) results for the relevant aliquoting and reading dates, no microbial growth was observed on personnel monitoring plates for analyst CCD on the date of testing (08Sep2026), the preceding sampling date (03Sep2026), or the subsequent sampling date (15Sep2026). Similarly, ISO 5 BSC (E001798) surface sampling and settling plates demonstrated no microbial growth on the date of testing (08Sep2026, CCD), the preceding sampling date (07Sep2026, ALA), or the subsequent sampling date (09Sep2026, ALA).

Weekly active air sampling of Cleanroom Suite 114 was performed for both the week of processing (04Sep2026) and the week of aliquoting (10Sep2026). On 04Sep2026 (performed by ISS), 1 CFU was recovered from the outermost ISO 8 anteroom (114) under ETX-260914-0487 (identification pending), while the intermediate ISO 7 buffer room (114A) and innermost ISO 7 cleanroom (114B) showed no microbial recovery (0 CFU). Similarly, on 10Sep2026 (performed by SMO), 1 CFU was recovered from the outermost ISO 8 anteroom (114) under ETX-260921-0520 (identification pending), with 0 CFU in both ISO 7 cleanrooms 114A and 114B. Weekly surface sampling across Suite 114 on both 04Sep2026 and 10Sep2026 demonstrated no microbial recovery (0 CFU) across all evaluated locations (Table 114, Cart 114, Cart 114A, Floor 114A, Cart 114B, Floor 114B).

All sample processing and aliquoting activities were performed within validated ISO 5 Biological Safety Cabinets (BSCs) E001316 and E001798, located in ISO 7 cleanrooms 114B and 114A, respectively. Furthermore, the test samples and media do not come into direct contact with the anteroom air or room surfaces at any stage of the testing process. Samples and media are prepared and transferred in disinfected, lidded containers and transported on carts within the controlled environments. Therefore, the single environmental monitoring recoveries from the outermost ISO 8 anteroom (114) originated from an area that is physically segregated from the controlled ISO 5 environments where the test sample was handled and processed.

The complete absence of microbial recovery from analyst glove plates, settling plates, and ISO 5 critical work surfaces throughout both processing and aliquoting indicates that no identifiable contamination transfer pathway existed between the room environment and the ISO 5 processing areas. Therefore, based on the absence of microbial recovery within the critical ISO 5 processing environments, satisfactory environmental monitoring results for personnel and critical surfaces, and the controlled processing conditions maintained throughout testing, the analyst or cleanroom environment is not considered a likely source of contamination for the sample.

All analysts confirmed full compliance with cleaning procedures as outlined in SOPs MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-44 (Celsis Sterility Testing). A review of the available data confirms that the cleanroom and equipment conditions remained within acceptable parameters. No deviations or obvious signs of laboratory error during processing and aliquoting were noted from analysts.

To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed on the same day. ETX-260828-0527 was the 1st sample processed in the batch by processing analyst - GS - on 01 Sep 2026. No other sample processed by GS on that date was found to be positive for microbial growth. A review of all other samples processed, incubated, aliquoted, and read within the same testing batch demonstrated no evidence of microbial growth. The absence of additional positive results within the batch confirms that cross-contamination between samples is unlikely during both the processing and the aliquoting steps.

An analysis of the six-month sample history for Optimal Balance Pharmacy (E19193) indicates that Eagle Analytical processed samples for Celsis Sterility testing with no prior occurrences of an out-of-specification or positive result for this analyte during this period.

A review of the lot history shows that there have been no additional submissions of sample lot: LG342010349 for retesting using Celsis Sterility Testing or any other sterility method.

Based on the observations outlined above, there is no clear indication that the contamination originated from reagents, supplies, the cleanroom environment, or the analyst. Therefore, the results are considered valid, and the likelihood of laboratory error is minimal.



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Has root cause been identified as a possible laboratory error?*Note: Root cause must be conclusive, verifiable, and well-documented.*☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☒ No, the OOS result is valid.**Reviewer comment(s):****Prepared by (Print):**

Qiyue Chen

Signature/Date:

SIGN

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

SIGN

Section E – Quality Disposition**Can investigation be closed?**☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.**Approved by QA Representative (Print):****Signature/Date:**

SIGN